

# CartFlow QA Suite

## Functional Requirements Specification

Frontend | Version 1.0

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| Project Name         | CartFlow QA Suite (Frontend)                                    |
| Client / Application | CartFlow Commerce                                               |
| Document Type        | Functional Requirements Specification                           |
| Version              | 1.0                                                             |
| Prepared By          | Purva Rawale                                                    |
| Creation Date        | April 01, 2026                                                  |
| Approval Date        | May 30, 2026                                                    |
| Reference URL        | <a href="https://demo.cartflow.in">https://demo.cartflow.in</a> |

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# 1. Purpose of the Document

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This document describes the functional requirements for the CartFlow QA Suite project — a manual testing project covering the frontend of a standard e-commerce web application. The purpose is to define what features are in scope for testing, what each feature is expected to do, and what the user should be able to accomplish.

This FRS acts as the foundation for writing test scenarios, test cases, and the RTM for this project. All test cases written for this project can be traced back to one or more requirements listed here.

This document is organised into four sections:

- Project Overview
- Functional Requirements (with Requirement IDs)
- Information Architecture (module-wise feature list)
- Sign-off

# 2. Project Overview

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CartFlow QA Suite is a manual frontend testing project done on an open-source e-commerce web application. The application allows online store owners to list products, manage inventory, and accept customer orders. End-users can register, browse products, add items to cart, and complete a purchase through a multi-step checkout.

For this project, the application was tested on its demo environment at <https://demo.cartflow.in> using manual testing techniques. The testing scope covers all major customer-facing modules — from account creation through to order completion.

## Technical Setup:

- Platform: PHP-based web application
- Database: MySQL
- Frontend: HTML, CSS, JavaScript
- Test Environment: Windows 10, Chrome (latest), Firefox (latest)
- Test Period: April 2026 – May 2026

# 3. Functional Requirements

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The table below lists all functional requirements identified from the application. Each requirement has a unique ID that maps to test scenarios and test cases in the RTM.

| Req ID | Module | Requirement Description | Priority |
|--------|--------|-------------------------|----------|
|--------|--------|-------------------------|----------|

|         |                 |                                                                                                                     |        |
|---------|-----------------|---------------------------------------------------------------------------------------------------------------------|--------|
| REQ-001 | Register        | User shall be able to create an account using First Name, Last Name, Email, Password, and accepting Privacy Policy. | High   |
| REQ-002 | Register        | System shall display a confirmation email to the registered email address after successful registration.            | High   |
| REQ-003 | Register        | System shall enforce password complexity — minimum 4 characters.                                                    | High   |
| REQ-004 | Login           | User shall be able to log in using a registered email and password.                                                 | High   |
| REQ-005 | Login           | System shall display a warning message for invalid or unrecognised credentials.                                     | High   |
| REQ-006 | Login           | System shall lock the account or show a warning after 5 consecutive failed login attempts.                          | High   |
| REQ-007 | Logout          | User shall be able to log out from 'My Account' dropdown or Right Column.                                           | High   |
| REQ-008 | Logout          | After logout, pressing browser back should not restore the session.                                                 | High   |
| REQ-009 | Forgot Password | User shall be able to request a password reset using their registered email address.                                | Medium |
| REQ-010 | Search          | User shall be able to search for products by name using the search bar.                                             | High   |
| REQ-011 | Search          | Search shall support filtering by category and subcategory.                                                         | Medium |
| REQ-012 | Product Display | Product page shall display image, product code, availability, price, quantity selector, and Add to Cart option.     | High   |
| REQ-013 | Add to Cart     | User shall be able to add products to cart from product page and featured product listing.                          | High   |
| REQ-014 | Shopping Cart   | Cart shall show product image, name, quantity, unit price, and total with options to update or remove items.        | High   |
| REQ-015 | Checkout        | System shall support a 6-step checkout: options, billing, delivery, delivery method, payment, confirm order.        | High   |
| REQ-016 | My Account      | Logged-in user shall be able to view and update account information, change password, and manage addresses.         | Medium |
| REQ-017 | Order History   | User shall be able to view past orders with details including product, quantity, price, and status.                 | Medium |
| REQ-018 | Wish List       | User shall be able to add products to a wish list and move them to cart.                                            | Low    |
| REQ-019 | Product Compare | User shall be able to compare up to 4 products side by side.                                                        | Low    |
| REQ-020 | Currency        | User shall be able to switch currency using the currency selector in the header.                                    | Medium |

## 4. Information Architecture

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### 4.1 Frontend Modules

The following sections describe the expected behaviour of each module tested in this project.

#### 4.1.1 Navigation

The application has a top-level category menu that displays parent product categories. Hovering over a parent category shows a dropdown with sub-categories. Clicking a category navigates to a filtered product listing page. The menu appears on all pages.

#### 4.1.2 Home Page

The home page is the default landing page. It includes a banner slideshow, featured product section, brand logos section, and footer. The home page gives the user quick access to product categories, search, and account options.

- Banner slideshow auto-rotates between banners and each banner is clickable
- Featured products section shows product image, name, price, and Add to Cart button
- Brand logos carousel is shown below featured products

#### 4.1.3 Header

The header is present on every page of the application and contains:

- Company logo — clicking it returns the user to the home page
- Currency selector — user can change the display currency
- Search bar — user can type a product name to search
- My Account dropdown — shows Register/Login options (or Account/Logout when logged in)
- Wish List, Shopping Cart, and Checkout quick links

#### 4.1.4 Product Display Page

Each product has its own page with:

- Main product image with thumbnail gallery for alternate views
- Product details: name, brand, product code, availability, and price
- Quantity selector and Add to Cart button
- Add to Wish List and Add to Compare options
- Description and Reviews tabs below the main section

#### 4.1.5 Category Listing Page

When a user clicks a category, they see a paginated list of products in that category. Products can be sorted by name, price, or rating. View can be toggled between grid and list. Pagination and Show (items per page) controls are present.

#### 4.1.6 Product Compare

The Add to Compare feature lets users select multiple products and view them side by side on a comparison page. The comparison table shows price, model, brand, availability, and specifications.

#### 4.1.7 Shopping Cart

When products are added, they appear in the shopping cart page. The cart shows:

- Product image, name, model, quantity (editable), unit price, and total

- Coupon code and gift certificate redemption fields
- Estimate Shipping and Tax section
- Continue Shopping and Checkout buttons

#### 4.1.8 Register and Login

New users can register by filling in First Name, Last Name, Email, Password, and agreeing to the Privacy Policy. Returning users log in with Email and Password. Guest checkout is available without registration.

- Registration sends a confirmation email to the registered address
- Login page shows Forgot Password link for password recovery
- Session should persist while browser is open and expire on logout

#### 4.1.9 Checkout (6 Steps)

After adding items to cart, the user proceeds through:

- Step 1: Checkout options — guest, login, or register
- Step 2: Billing details — name, email, address
- Step 3: Delivery details — same as billing or different address
- Step 4: Delivery method — select shipping option
- Step 5: Payment method — select payment option
- Step 6: Confirm order — order summary, then confirm

After confirmation, a success message is shown and the order appears in Order History.

## 5. Sign-Off

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The parties below have reviewed and agreed to the requirements defined in this document. Any significant changes after approval will require a review of existing test cases and project scope.

| Role             | Name                                | Date         |
|------------------|-------------------------------------|--------------|
| QA Tester (Solo) | Purva Rawale                        | May 30, 2026 |
| Business Lead    | <i>Self-reviewed (solo project)</i> | May 30, 2026 |
| Project Manager  | <i>Self (solo project)</i>          | May 30, 2026 |

— End of FRS Document SS—